

EX NO: 6 Transfer Files Between Virtual Machines

Aim:

To understand and perform file transfer between virtual machines (VMs) using different methods like **SCP**, **Shared Folders**, and **FTP**

Requirements:

- Two Virtual Machines (VMs) running (Ubuntu recommended)
- Network connectivity between both VMs
- Installed tools like `ssh`, `scp`, or an FTP server

Procedure :

Method 1: Using SCP (Secure Copy Protocol)

Step:

1. **Ensure SSH is installed and enabled on both VMs:**
2. `sudo apt update`
3. `sudo apt install openssh-server -y`
4. `sudo systemctl start ssh`
5. `sudo systemctl enable ssh`
6. **Find the IP address of the target VM:**
7. `ip a`

Example: 192.168.56.102

8. **Transfer file from VM1 to VM2 using SCP:** On VM1, run:
9. `scp /path/to/file.txt user@192.168.56.102:/home/user/`
 - o Replace `/path/to/file.txt` with the actual path.
 - o Replace `user` with the username of VM2.
10. **Enter the password of the target VM when prompted.**
11. **Verify** that the file has been transferred by logging into VM2 and checking the path.

Method 2: Using Shared Folder in VirtualBox

Step:

1. **Install Guest Additions on both VMs:** In VirtualBox menu: `Devices` → `Insert Guest Additions CD Image...`

Inside VM:

```
sudo apt install build-essential dkms linux-headers-$(uname -r)
sudo mount /dev/cdrom /mnt
sudo /mnt/VBoxLinuxAdditions.run
```

2. **Create a shared folder on the host:**
 - o Go to VM settings → Shared Folders → Add New Shared Folder.
 - o Select a folder path and name it (e.g., SharedVM).
 - o Check "Auto-mount" and "Make Permanent".
3. **Access shared folder from VM:** The folder is usually mounted at
/media/sf_SharedVM/.

Method 3: Using FTP Server

Step:

1. **Install vsftpd (FTP server) on VM2:**
2. `sudo apt update`
3. `sudo apt install vsftpd -y`
4. `sudo systemctl start vsftpd`
5. `sudo systemctl enable vsftpd`
6. **Transfer file using FTP from VM1:**
7. `ftp 192.168.56.102`
 - o Login with VM2 credentials
 - o Use `put filename.txt` to upload
 - o Use `get filename.txt` to download
8. **Exit FTP session:**
9. `bye`

Alternative Method: Using Remote Desktop (for Windows VMs)

- Enable RDP on both Windows VMs.
- Use drag-and-drop or clipboard sharing via RDP session.

Result:

Successfully transferred files between virtual machines using SCP, Shared Folders, and FTP methods.

Viva Questions:

1. What is SCP and how does it ensure secure file transfer?
2. What is the difference between SCP and FTP?
3. How are shared folders configured in VirtualBox?
4. What is the role of SSH in SCP?
5. Mention the advantages of using SCP over FTP.